

IS Practice

September 20, 2026

Question: CBC-MAC

A sender uses CBC-MAC with a 4-bit block cipher. The initialization vector (IV) is set to

$$IV = 0000.$$

The message consists of the following three 4-bit blocks:

$$M_1 = 1100, \quad M_2 = 1010, \quad M_3 = 0110.$$

For this key K , the required block-cipher encryption outputs are given below:

X	$E_K(X)$
1100	0011
1001	1110
1000	0101

CBC-MAC is computed as

$$C_0 = IV,$$

$$C_i = E_K(M_i \oplus C_{i-1}), \quad i = 1, 2, 3.$$

1. Compute C_1 , C_2 , and C_3 . Show the XOR operation at each step.
2. What is the final CBC-MAC value for the message?
3. Briefly explain why the final CBC-MAC value depends on all three message blocks.

Question: HMAC

A simplified HMAC system uses a hash function H with a block size of $b = 8$ bits. The secret key and message are

$$K = 1011, \quad m = 11001010.$$

The HMAC constants are

$$ipad = 00110110, \quad opad = 01011100.$$

Since the key is shorter than the block size, it is right-padded with zeros to form the 8-bit key K_0 .

The required hash outputs are given as

$$H((K_0 \oplus ipad) || m) = 0110$$

and

$$H((K_0 \oplus opad) || 0110) = 1011.$$

Recall that

$$\text{HMAC}(m, K) = H((K_0 \oplus opad) || H((K_0 \oplus ipad) || m)).$$

1. Find the padded key K_0 .
2. Compute $K_0 \oplus ipad$ and $K_0 \oplus opad$.
3. Identify the intermediate (inner) hash value.
4. Compute the final HMAC value.
5. Briefly explain why HMAC uses two hashing stages with different constants, $ipad$ and $opad$.

Question: Digital Signatures

Bob wants to send a digitally signed message M to Alice.

For the message M , the cryptographic hash function produces

$$H(M) = 1011.$$

Bob signs this hash using his private key. The digital signature generation algorithm gives

$$S = \text{Sign}_{K_{\text{priv}}}(1011) = 0110.$$

Bob sends the following to Alice:

$$(M, S).$$

When Alice applies Bob's public key to the received signature, the signature verification algorithm recovers

$$\text{Verify}_{K_{\text{pub}}}(0110) = 1011.$$

1. What hash value does Bob sign?
2. What digital signature does Bob generate?
3. What two items does Bob send to Alice?
4. Explain how Alice verifies the digital signature. Is the signature valid?
5. Suppose an attacker modifies the message to M' such that

$$H(M') = 1001.$$

The signature $S = 0110$ remains unchanged. Will Alice accept the signature? Briefly justify your answer.

Question: RSA

Consider an RSA system with the following prime numbers:

$$p = 11, \quad q = 7.$$

Let the public exponent be

$$e = 7.$$

1. Compute the modulus

$$n = p \times q.$$

2. Compute Euler's totient

$$\phi(n) = (p - 1)(q - 1).$$

3. Verify that $e = 7$ is a valid choice by checking that

$$\gcd(e, \phi(n)) = 1.$$

4. Find the private exponent d such that

$$d \times e \equiv 1 \pmod{\phi(n)}.$$

5. Write the RSA public key (e, n) and private key (d, n) .

6. If the plaintext message is

$$M = 4,$$

encrypt it using

$$C = M^e \bmod n.$$

7. Decrypt the ciphertext using

$$M = C^d \bmod n.$$

Verify that the original plaintext is recovered.

Question: Diffie–Hellman and MITM

Alice and Bob publicly agree on

$$p = 23, \quad g = 5.$$

Alice chooses $x = 4$ and Bob chooses $y = 3$.

1. Compute their public values:

$$A = g^x \bmod p, \quad B = g^y \bmod p.$$

2. Compute the shared secret from both sides:

$$K_A = B^x \bmod p, \quad K_B = A^y \bmod p.$$

Verify that both values are equal.

3. If an attacker knows p , g , A , and B , what mathematical problem must be solved to recover x or y ?
4. Suppose Mallory performs a man-in-the-middle attack using private value $m = 6$ and public value

$$M = g^m \bmod p.$$

Mallory replaces Alice's and Bob's public values with M . Compute:

$$M, \quad K_{AM} = M^x \bmod p, \quad K_{BM} = M^y \bmod p.$$

Do Alice and Bob still share the same key? Briefly explain how Mallory can intercept their communication.

Question: Modular Exponentiation

Compute

$$29^{11} \bmod 53$$

without directly calculating 29^{11} .

Use exponent decomposition and the modular reduction property

$$(a \cdot b) \bmod m = ((a \bmod m)(b \bmod m)) \bmod m.$$

1. Express 11 as a sum of suitable powers.
2. Compute the required intermediate powers modulo 53.
3. Find the final value of $29^{11} \bmod 53$.

Question: Modular Exponentiation

Compute

$$37^{13} \bmod 61$$

without directly calculating 37^{13} .

Use exponent decomposition and modular reduction to simplify the calculation.

1. Express 13 as a sum of suitable powers.
2. Compute the required intermediate powers modulo 61.
3. Find the final value of $37^{13} \bmod 61$.

Question: CBC Mode Encryption

A message is divided into three 4-bit plaintext blocks:

$$P_1 = 1010, \quad P_2 = 1100, \quad P_3 = 0111$$

and the initialization vector is

$$IV = 0011.$$

CBC encryption uses

$$C_i = E_K(P_i \oplus C_{i-1}), \quad C_0 = IV.$$

For the given key K , the required block-cipher outputs are:

X	$E_K(X)$
1001	0101
1001	0101
0010	1110

1. Compute C_1 , C_2 , and C_3 .
2. Show the XOR operation before each encryption.
3. Briefly explain why changing P_1 can affect all later ciphertext blocks.

Question: Kasiski's Method

Consider the following Vigenère ciphertext:

QWEABCQWEDEFABC

Using Kasiski's method:

1. Identify suitable repeated sequences in the ciphertext.
2. Determine the positions and distances between their occurrences.
3. Find the common factors of the distances.
4. Use the GCD of the distances to estimate the most likely key length.

Question: Kasiski's Method

Consider the following Vigenère ciphertext:

QWEABCXYQWERSTUABC

Using Kasiski's method:

1. Identify the repeated sequences in the ciphertext.
2. Determine their positions and the distances between repetitions.
3. Find the common factors of the distances.
4. Use the GCD of the distances to estimate the most likely key length.